

DW_FamilyFinance – SSIS ETL Documentation

Version 1.0

SQL Server Integration Services (SSIS) 2022

Project Information

Project Name: DW_FamilyFinance

Solution Name:

DW_FamilyFinance

SSIS Version:

SQL Server Integration Services 2022

Development Tool:

Visual Studio 2022
SSIS Projects Extension

Database:

STG_FamilyLiving

Purpose:

Develop an automated ETL framework that loads Family Finance source files from OneDrive into SQL Server staging tables while maintaining auditability, metadata tracking, and duplicate load prevention.

Current SSIS Packages

1. STG_LoadExpenseData.dtsx

Purpose:

Load expense transactions from Citi Bank and Wells Fargo Excel files into STG_FamilySourceData.

Current Status:

Completed

2. STG_LoadIncomeData.dtsx

Purpose:

Load payroll and income data into Family_Income staging table.

Current Status:

Initial Package Completed

3. STG_LoadDescriptionData.dtsx

Purpose:

Load description lookup data into STG_Description_LookUp.

Current Status:

Completed

Expense ETL Package Design

Package Name:

STG_LoadExpenseData.dtsx

Control Flow

Current Design

```
FELC_LoadExpenseFiles
|
|— ST_ParseExpenseFileMetadata
|
|— EST_CheckFileAlreadyLoaded
|
|— EST_DeleteOldExpenseFileRows
|
|— DFT_LoadExpenseData
|
|— EST_InsertFileLoadAudit
```

Foreach Loop Container

Container Name:

FELC_LoadExpenseFiles

Purpose:

Loop through all expense Excel files located in the OneDrive expense folder.

Folder:

C:\Users\tbeha\OneDrive - Behailu Tessema-(DeveloperMode.dev)
\FamilyFinance\STG_FamilyLiving_Expense

File Pattern:

*.xlsx

Variables

User Variables

Variable	Data Type	Purpose
FullFilePath	String	Full Excel file path
FileName	String	Current Excel file name
SourceName	String	Citi or WellsFargo
FileYear	Int32	Year extracted from file
RowCount	Int32	Number of rows loaded
FileAlreadyLoaded	Boolean	Incremental load control
SourceFolder	String	OneDrive source folder

Dynamic Excel Connection

Connection Manager:

STG_Expense_OneDriveSource

Property Expression:

ExcelFilePath

Expression:

```
@[User::FullFilePath]
```

Purpose:

```
Dynamically switch Excel source file during Foreach Loop processing.
```

Script Task

Task Name:

```
ST_ParseExpenseFileMetadata
```

Language:

```
Microsoft Visual Basic 2022
```

Purpose:

```
Extract metadata from current file name.
```

Outputs:

```
User::FileName  
User::SourceName  
User::FileYear
```

Examples:

```
Citi_YTD_2026.xlsx  
  SourceName = Citi  
  FileYear = 2026  
  
WellsFargo_2026_YTD.xlsx  
  SourceName = WellsFargo  
  FileYear = 2026
```

Data Flow

Task Name:

DFT_LoadExpenseData

Flow:

EXL_ExpenseSource
↓
DC_AddFileMetadata
↓
RC_ExpenseRowCount
↓
OLED_STG_FamilySourceData

Derived Column Transformation

Transformation:

DC_AddFileMetadata

Added Columns:

FileName

Expression:

(DT_WSTR,255)UPPER(@[User::FileName])

FileYear

Expression:

@[User::FileYear]

SourceName

Expression:

```
(DT_WSTR,50)(UPPER(@[User::SourceName]) + " BANK")
```

Examples:

```
CITI BANK  
WELLSFARGO BANK
```

Row Count Transformation

Transformation:

```
RC_ExpenseRowCount
```

Variable:

```
User::RowCount
```

Purpose:

```
Capture number of rows loaded from current Excel file.
```

Destination Table

Table:

```
STG_FamilyLiving.dbo.STG_FamilySourceData
```

Metadata Columns Added:

FileName
FileYear
SourceName

Incremental Load Framework

Task:

EST_CheckFileAlreadyLoaded

Purpose:

Prevent duplicate file loads.

SQL:

```
SELECT
    CASE
        WHEN EXISTS
            (
                SELECT 1
                FROM STG_FamilyLiving.dbo.STG_FileLoadAudit
                WHERE UPPER(FileName) = UPPER(?)
                AND LoadStatus = 'SUCCESS'
            )
        THEN CAST(1 AS BIT)
        ELSE CAST(0 AS BIT)
    END AS FileAlreadyLoaded;
```

Variable:

User::FileAlreadyLoaded

Precedence Constraint

Expression:


```
@[User::FileAlreadyLoaded] == False
```

Behavior:

```
TRUE  = Load file  
FALSE = Skip file
```

File Load Audit

Audit Table:

```
STG_FamilyLiving.dbo.STG_FileLoadAudit
```

Columns:

```
FileLoadAuditID  
PackageName  
FileName  
FilePath  
SourceName  
FileYear  
SourceRowCount  
LoadStatus  
LoadDateTime  
LoadedBy
```

Audit Insert Task

Task:

```
EST_InsertFileLoadAudit
```

Purpose:

Record successful file loads.

Values Captured:

Package Name
File Name
File Path
Source Name
File Year
Source Row Count
Load Status
Load Date Time
Loaded By

Current Audit Results

Successfully Loaded:

Citi_YTD_2020.xlsx
Citi_YTD_2021.xlsx
Citi_YTD_2022.xlsx
Citi_YTD_2023.xlsx
Citi_YTD_2024.xlsx
Citi_YTD_2025.xlsx
Citi_YTD_2026.xlsx

WellsFargo_2020_YTD.xlsx
WellsFargo_2021_YTD.xlsx
WellsFargo_2022_YTD.xlsx
WellsFargo_2023_YTD.xlsx
WellsFargo_2024_YTD.xlsx
WellsFargo_2025_YTD.xlsx
WellsFargo_2026_YTD.xlsx

Total Files Audited:

14

Current Package Status

Expense Package:

Completed Version 1.0

Features:

- ✓ Dynamic Excel Processing
- ✓ OneDrive Integration
- ✓ Metadata Extraction
- ✓ FileName Tracking
- ✓ FileYear Tracking
- ✓ SourceName Tracking
- ✓ Audit Framework
- ✓ Duplicate File Prevention
- ✓ Row Count Capture
- ✓ Automated File Loop Processing

Future Enhancements

Version 2.0

FileName + SourceRowCount Incremental Logic

If FileName Exists
 AND SourceRowCount Unchanged
 Skip

If FileName Exists
 AND SourceRowCount Changed
 Delete Existing Rows
 Reload Updated File
 Insert New Audit Record

Purpose:

Support YTD files that grow monthly.